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Technical Partner DELL Technologies

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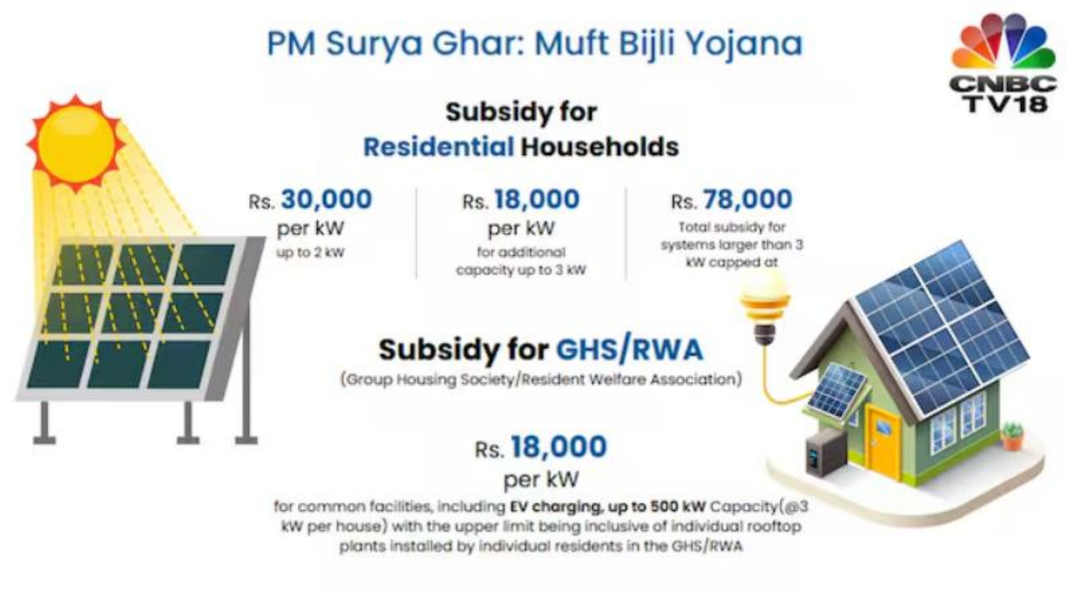
Problem Statement: Intelligent Solar Feasibility and Planning System for Buildings.

Problem Statement

- Solar energy adoption processes are highly complex and fragmented, lacking an automatic, accessible way for property owners to assess their feasibility
- Homeowners and building managers struggle to get accurate, personalized system sizing and realistic financial payback data without specialist consultations
- Results in widespread consumer hesitation, prolonged decision-making, and a drastically slowed transition to renewable energy



Statistic

- Studies indicate that manual feasibility quoting and customer acquisition can account for up to 50-60% of the total price of a residential solar installation, acting as a major artificial barrier to widespread adoption

Proposed Solution

- An AI-powered solar feasibility platform that automatically analyzes a building's historical electricity consumption data and generates a fully personalized solar installation plan — eliminating the need for expert consultation.
- Ingests detailed electricity usage patterns across time-of-day and seasonal cycles to accurately determine the optimal solar capacity for partial or complete grid replacement.
- Integrates solar irradiance, geographic, and weather datasets to predict real-world energy generation potential based on the building's location, orientation, and environmental conditions.
- Delivers a comprehensive financial blueprint, including panel requirements, rooftop space utilization, installation costs, maintenance estimates, and payback period — all computed automatically from real data.
- Provides interactive dashboards showcasing energy consumption trends, projected solar output, CO_2 emission reductions, and financial returns, enabling users to make informed, data-driven decisions at a glance.

Technology Stack

Programming Languages
- Python
- JavaScript
- TypeScript

Models / Logic
- Meta Prophet
- XGBoost
- Custom Solar Sizing Algorithm

Frameworks and Libraries
- React.js
- Tailwind CSS
- FastAPI
- Recharts
- React Leaflet
- PVLIB (Solar Energy Simulation)

Database / Storage
- MongoDB

Methodology

Steps:

- Data Ingestion & Building Profile
- Consumption Analysis & Forecasting
- Solar Potential Estimation
- Optimal System Sizing
- Financial Viability Report
- AI-Powered Advisory & Dashboard

Data ingestion & building profile → Commercial Analysis & Forecasting → Solar Potential Estimation → System Sizing & Optimization → Financial Viability report → AI Advisory and Dashboard

Impact:

Who Benefits:

- Home owners and building owners
- Environment and local communities
- Solar installer and policymakers

What Change It Brings:

- Simplifies complex solar feasibility assessment
- Promotes transition to renewable energy
- Accelerates decision-making for solar investments
- Reduces carbon footprints of buildings

Example

- Provides property owners with instant, easy-to-understand system sizing and financial payback data without needing expert consultations
- Drives the shift toward clean energy adoption, reducing the community's dependency on traditional power grids
- Turns a lengthy planning process into a quick upload-and-view experience, making solar adoption accessible to everyone
- Offers tangible metrics on CO_2 offsets (like equivalent trees planted), allowing users to see their direct environmental contribution

Feasibility & Innovation

Feasibility

- Uses a lightweight web-based application accessible directly from standard desktop and mobile browsers
- Requires only basic, easily accessible inputs, seamlessly processing standard electricity bill images or PDFs
- Integrates reliably with established global datasets for precise solar irradiance
- Built using highly scalable, industry-standard modern web technologies

Innovation

- Combines AI-powered OCR bill parsing with dynamic, custom solar sizing algorithms
- Uses interactive geospatial mapping to pinpoint exact building coordinates for localized solar data
- Integrates immediate financial forecasting alongside clear, tangible environmental metrics (like CO_2 offset)
- Creates a unified, end-to-end platform transitioning users from basic utility data to a complete solar action plan

Example

- Works smoothly on everyday smartphones, enabling homeowners to simply snap a photo of their electricity bill to get started
- Generates system sizing and recommendations based on the user's actual extracted consumption data rather than generic estimates
- Empowers property owners by translating complex solar feasibility math into interactive, easy-to-understand visual dashboards

Future Scope

Scaling Possibilities

- Expansion to commercial and industrial sectors alongside residential deployment
- Integration with regional utility boards and national solar subsidy schemes

Extensions

- Advanced AI-based models for predicting future household energy needs
- Predictive analytics for precise battery storage and inverter sizing
- Dedicated mobile app connecting homeowners directly with verified local solar installers

Example

- Real-time matching of high-intent homeowners with available local solar contractors
- Automated detection and application of eligible government rebates and tax incentives
- Integration with smart home platforms and EV (Electric Vehicle) charging ecosystems to size systems for future usage

References

- Ministry of New and Renewable Energy (MNRE) – Rooftop Solar Program Guidelines and Reports
- NITI Aayog – Renewable Energy Integration & Transition Planning Reports
- International Energy Agency (IEA) – Global Solar Energy Market & Feasibility Studies
- National Renewable Energy Laboratory (NREL) – PV System Performance and Irradiance Methodologies

https://www.niti.gov.in/sites/default/files/2021-07/Renewable-Energy-Integration-in-India-2021.pdf

Thank You